Cynthia Chen

Seattle, WA
(732)-425-4289
cynthiachen070@gmail.com

github: cynthiachennn

linkedin: Cynthia Chen

Education

Rutgers University - New Brunswick - *B.S. Computer Science, Cognitive Science (Minds, Machines and Computation track)*

SEP 2021 - MAY 2022
New Brunswick, NJ

- GPA: 3.95
- **Awards:**
  - Dean's List, Graduated with Honors, Summa Cum Laude

Research Experience

Allen Institute for Data Driven Discovery in Biology
Postbac Research Fellow - Cindy Poo and Michael Buice

JUN 2024 - PRESENT
Seattle, WA

- Designed a neural network architecture that incorporated classical Reinforcement Learning algorithms to solve problems of decision making under uncertainty, outperforming other models
- Utilized cloud computing and GPU acceleration techniques to train deep learning models on large amounts of data
- Presented the emergence of interpretable value representations in novel machine learning architecture at AAAI Conference
- Designed Markov models to represent state-mediated decision making and belief update processes observed in rodent experiments
- Used fitting techniques such as regression and likelihood estimation to model decision making strategies that predict experimental data

NeuroCognition Lab
NSF REU Research Assistant - Adham Atyabi

JUN 2023 - NOV 2023
Colorado Springs, CO

- Applied data processing and feature extraction techniques on EEG data such as Fourier transforms and Independent Component Analysis with EEGLab
- Designed Convolutional Neural Networks and Transformer models for EEG decoding for Brain Computer Interface applications that competed with the state of the art models in classification accuracy
- Presented research and findings to REU cohort students and authored a symposium paper detailing experimental methodology and results

Social Entrepreneurship, Networks, Digital Innovation Lab
Aresty Summer Science Research Assistant - Sophia Fu

JUN 2021 - MAY 2023
New Brunswick, NJ

- Awarded Aresty Summer Science grant to work on research project and presented poster at corresponding Symposium
- Conducted exploratory analysis and literature review for abstract presented at International Communication Association Conference
- Analyzed and coded 50+ papers and interview transcripts about nonprofit resilience and stakeholder management
- Evaluated BERTtopic classifications of twitter data on climate change and mental health

Skills

Technical Skills - python, java, C, MATLAB/EEGLAB, Git, Docker, Pytorch (CUDA), Cloud Computing
Machine Learning - CNNs, Transformers, Reinforcement Learning, Markov Models, Regression
Scientific Communication - Scientific writing, Oral Presentations, Poster design and presention

Publications and Presentations

Chen, C., Johnson, S., Buice, M., & Khalvati, K. (2025). "Information-based Value Iteration Networks for Decision Making Under Uncertainty" *International Conference on Learning Representations (ICLR) 2026* **[under review]**

Chen, C., Buice, M., & Khalvati, K. (2025). "Factorized Value Iteration Network for Decision Making under Mixed Observability." *Generalization in Planning Workshop, AAAI 2025*.

Lin, H., Zhang, R., Scrivani, K., **Chen, C.**, Fu, S.,(2024). "Mapping the Landscape of Online Discourse on Climate Change-displaced Populations: A Topic Modeling Analysis with BERTopic." *74th Annual ICA Conference*

Chen, C., & Atyabi, A. (2023)." Finger Flex Classification for Brain Computer Interfaces." *REU Symposium on Deep Learning.*

Relevant Community Service

BrainFest - Pacific Science Center
NOV 2024

Science Booth Coordinator

- Facilitated interactive booth teaching visitors mainly aged 5-15 about EMG signals and technologies that utilize them

Rutgers Cognitive Science Club
SEP 2021 - MAY 2024

Executive Board, Graphic Design

- Designed graphics to promote weekly events
- Assisted marketing team to ensure events are properly represented in promotional material
- Collaborated with executive board members and external organizations to plan and organize weekly club events
- Researched and created interactive presentations on topics in cognitive science to general club members

Rutgers Undergraduate Research Journal
SEP 2021 - MAY 2022

Peer reviewer

- Contributed to intensive peer review process of three papers for an interdisciplinary undergraduate journal
- Collaborated with editorial committee on curation of accepted papers for the journal issues